Anubhav Das

Date of Birth - 06/08/2004
Sex - Male
Nationality - Indian
Languages - English, Hindi, Bengali

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Rajiv Gandhi Institute of Petroleum Technology
(Institute of National Importance)

Education

Rajiv Gandhi Institute Of Petroleum Technology — *Jais, Amethi*
B.Tech. in Mathematics and Computing (CGPA: 7.54 / 10) — 2023-27

Senior Secondary — *Percentage - 88.2*
Pioneer Convent School, Indore, CBSE Board — Academic Year - 2022

Secondary — *Percentage - 88.2*
Servite Convent School, Narmadapuram, CBSE Board — Academic Year - 2020

Internship Experience

India Meteorological Department (IMD) — *New Delhi, India*
Research Intern — Numerical Weather Prediction (NWP) & Deep Learning — Summer 2025

- Worked on improving existing functional NWP models used operationally for weather forecasting at IMD.
- Modified a U-Net architecture (deviating from the traditional encoder-decoder design) and integrated a SWIN Transformer, further architecting the hybrid model for **Thunderstorm prediction**.
- Designed and implemented a hybrid **MTGNN (Multivariate Time-Series Graph Neural Network)–U-Net** architecture for **short-term Sandstorm prediction**, replacing the traditional NCUM model that IMD had been using operationally for years.

Projects

Weather Forecasting U-Net
Python, PyTorch, NOAA Reanalysis Data, NetCDF — GitHub

- Built a 4-level U-Net with skip connections to forecast tornado and significant-tornado probability heatmaps from NOAA atmospheric reanalysis fields (CAPE, CIN, Geopotential Height).
- Implemented custom preprocessing pipeline to extract daily 2D slices from NetCDF data, interpolate to 256×256, and apply weighted BCE loss for rare-event class imbalance.
- Created inference module with NOAA SPC-standard colormaps for historically accurate probability visualisation and bulk evaluation of severe weather days.

MTGNN–U-Net Hybrid Architecture
Python, PyTorch, Graph Neural Networks, Spatio-Temporal Modelling — GitHub

- Architected a hybrid deep learning model combining a U-Net spatial encoder/decoder with an MTGNN (Multivariate Time-Series Graph Neural Network) bottleneck for multi-day tornado probability forecasting.
- MTGNN bottleneck learns directed graph connections between 256 spatial nodes (16×16 grid) over 7-day input sequences using dilated temporal inception layers and mix-hop graph convolutions.
- Designed the model to ingest 7-day atmospheric sequences and output 3-day-ahead continuous probability heatmaps, replacing sequential-day independent inference.

Support-AI
FastAPI, PostgreSQL, pgvector, OpenAI, SQLAlchemy, Docker — GitHub

- Developed a RAG-based AI support system with document upload, parsing (PDF), semantic chunking, and vector-embedding storage using pgvector for similarity search.
- Built a production-grade FastAPI backend with async PostgreSQL, Alembic migrations, JWT authentication, structured logging (structlog), and background document processing pipelines.
- Containerised the full stack with Docker Compose and implemented comprehensive test coverage with pytest-asyncio.

Achievements

IIT Bombay NEC finalist Top 50 out of 500+ colleges for excellence in campus entrepreneurship initiatives ,represented our college in IIT Bombay.

Indian Delegate – AYIMUN 13th, Kuala Lumpur Selected to represent India for international youth diplomacy and leadership conference held in Kuala Lumpur .

IIT kharagpur LSM EAD finalist Finalist in the Prestigious Entrepreneurial and Development competition by IIT kharagpur

Positions of Responsibility

Head Entrepreneurship Cell, RGIPT — *August 2025 - Present*
Led a team of 96 members, organized several events, Partnered with several startups for various events

Coordinator Publicity and PR Coordinator of RGIPT'S Annual Fest week - URJA SANGAM — *August 2025 - Present*
Coordinated PR and outreach for URJA SANGAM fest, reaching 3M+ viewership online and 30+ colleges

Social Media Head Entrepreneurship Cell, RGIPT — *February 2025 - August 2025*

Video Editing Head Entrepreneurship Cell, RGIPT — *February 2025 - August 2025*

DevClub Member, Geeks For Geeks RGIPT — *October 2023 - Present*

Event Management Head Society for Industrial Applied Mathematics RGIPT — *July 2024 - December 2024*

Technical Skills

Programming: Python, C++, C, JavaScript

Business & Sales: Lead Generation, Pipeline Management, Competitive Analysis, Cold Outreach, Market Research

AI & Productivity: Claude (Agentic Workflows), ChatGPT, Ollama, Prompt Engineering, Documentation, Automation

Tools & OS: Claude Code, VS Code, Postman, Jupyter Notebook, Anaconda, Figma

Design & Growth Tools: Figma, Canva, LinkedIn Sales Navigator, Apollo.io, HubSpot, GitHub

Web Skills: HTML5, CSS3, JavaScript, ReactJS, NextJS, Node, Express.js, MongoDB, REST APIs